Moral Hazard with Dual Risk-Averse Agent*

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Abstract

This note studies a stylized moral hazard problem with a single agent who is protected by the limited liability constraint and is *dual risk-averse* in the sense of Yaari (1987). I consider the principal's cost-minimizing problem to induce a given action. Under some regularity conditions, I show that the solution to a relaxed problem where the agent's incentive compatibility constraint is replaced by its first-order condition is a *(debt and) capped bonus* contract—the slope of wage is 0 for (low- and) high-outcomes and is 1 elsewhere. I also show that the first-order approach is valid if the outcome distribution is (in proportionate terms) less convex in action than the disutility of action, which is implied by Rogerson (1985)'s convexity of distribution condition and convex disutility.

1 Introduction

Since the seminal work of Holmström (1979), a large literature on moral hazard has studied how a principal can incentivize an agent to take desirable actions when the agent's action is not contractible. A persistent critique of the theory is that it often predicts optimal contracts that are complex and highly sensitive to the underlying distribution of outcomes, in contrast to the simple contracts frequently observed in practice.

This note offers an alternative rationale for such simple linear contracts. I consider a setting in which the agent is *dual risk-averse* in the sense of Yaari (1987); that is, he overweights the

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likelihood of "bad" outcomes when making decisions. I show that the cost-minimizing contracts for the principal take the form of a class of linear contracts, which I dub *debt and capped bonus contracts*. These contracts feature flat regions (i.e., slope zero) for low outcomes (debt) and high outcomes (capped bonus), and a slope of one in between—hence the name. This note contributes to the literature on moral hazard by providing a novel rationale (dual risk aversion) for the simple contracts observed in practice (for a "traditional" rationale, see Holmström and Milgrom (1987), among others).

The remainder of the note proceeds as follows. Section 2 introduces the model. Section 3 presents the main results. Section 4 concludes. The proof of Proposition 1 is relegated to Appendix A.

2 Model

Environment: I consider a stylized moral hazard environment with a single principal and a single agent. The agent takes a hidden action $a \in [0, \overline{a}]$ at a disutility $\psi(a)$, which is strictly increasing and C^2 with $\psi(0) = 0$. I assume that ψ satisfies the Inada conditions: $\psi'(0) = 0$ and $\psi'(\overline{a}) = +\infty$. The agent's action affects the distribution of outcome x, which takes a value in $\left[\underline{x}, \overline{x}\right]$. Given action a, the outcome x has a distribution $F(x \mid a)$ with everywhere positive density $f(x \mid a)$. I assume that F and f are C^2 and that the agent's action produces a first-order stochastic dominant shift on the outcome; $F_a(x \mid a) < 0$ for all $x \in \left(\underline{x}, \overline{x}\right)$ and all a.[1]

Preferences: The principal offers a bounded wage schedule $w\colon \left[\underline{x}, \overline{x}\right] \to [-m, M]$ to the agent, where $m,\ M \in \mathbb{R}_+$ and M is sufficiently large. She is risk-neutral and maximizes the expected value of her net benefit $x - w(x)$. The agent is *dual risk-averse* in the sense of Yaari (1987): given a distribution of wage $\Gamma^w(\cdot \mid a)$ induced by wage schedule w and action a, his gross utility is given by

$$\int_{-m}^{M} \phi(1 - \Gamma^w(v \mid a))\, \mathrm{d}v, \tag{2.1}$$

[1]Note that $F_a\left(\underline{x} \mid a\right) = F_a(\overline{x} \mid a) = 0$ for all a since the support of the outcome is fixed.

where distortion function $\phi\colon [0,1] \to [0,1]$ is defined as

$$\phi(p) = \begin{cases} 0 & \text{if } 0 \le p \le q \\ \frac{p-q}{1-q} & \text{if } q < p \le 1 \end{cases}$$

for some $q \in (0,1)$.[2]

Admissible wage schedules: I assume $m = 0$ so that the agent is protected by the limited liability constraint. Following Poblete and Spulber (2012), I restrict attention to wage schedules that satisfy the following assumptions.

Assumption 1 (Monotonicity)**.** $w(x)$ is nondecreasing.

Assumption 2 (Bounded slope)**.** $x - w(x)$ is nondecreasing.

Remark 1. Assumption 1 prevents the agent from sabotaging the task while Assumption 2 prevents the principal from doing so after the action has been chosen.

Assumptions 1 and 2 together imply that w is almost everywhere differentiable and Lipschitz continuous with

$$0 \le w'(x) \le 1 \text{ for almost every } x. \tag{Lipschitz}$$

3 Cost-minimizing wage schedules

3.1 Principal's problem

Suppose that the principal wants to induce action $a > 0$. Given Assumption 1, the distribution of wage is given by $\Gamma^w(v \mid a) = F\big(w^{-1}(v) \mid a\big)$, where w^{-1} is the generalized inverse of w:

$$w^{-1}(v) = \sup\{x\colon w(x) \le v\}.$$

[2] This is a special case of dual risk-averse preference. Yaari (1987) characterizes the dual representation for general continuous and nondecreasing ϕ satisfying $\phi(0) = 0$ and $\phi(1) = 1$. He also characterizes risk aversion by the convexity of ϕ.

Let $x(a) \equiv F^{-1}(1-q \mid a)$. The agent's gross utility (2.1) can be written as

$$\begin{aligned}\int_0^M \phi\Big(1 - F\big(w^{-1}(v) \mid a\big)\Big)\,\mathrm{d}v &= w(\underline{x}) + \int_{\underline{x}}^{\overline{x}} \phi(1 - F(x \mid a))\, w'(x)\,\mathrm{d}x \\ &= w(\underline{x}) + \int_{\underline{x}}^{x(a)} \frac{1 - F(x \mid a) - q}{1-q}\, w'(x)\,\mathrm{d}x,\end{aligned}$$

where the first equality follows from the change of variable $x = w^{-1}(v)$. For the agent to accept wage schedule w and choose action a, his net utility $U(w,a)$ must be greater than or equal to the nonnegative reservation utility U_0:[3]

$$\begin{aligned} & U(w,a) \equiv w(\underline{x}) + \int_{\underline{x}}^{x(a)} \frac{1 - F(x \mid a) - q}{1-q}\, w'(x)\,\mathrm{d}x - \psi(a) \geq U_0 \\ \iff & \int_{\underline{x}}^{x(a)} \frac{f(x \mid a)}{1-q}\, w(x)\,\mathrm{d}x - \psi(a) \geq U_0. \end{aligned} \tag{PC}$$

He chooses action a if it maximizes $U(w,\tilde{a})$:

$$a \in \operatorname*{arg\,max}_{\tilde{a}\in[0,\overline{a}]} \left[w(\underline{x}) + \int_{\underline{x}}^{x(\tilde{a})} \frac{1 - F(x \mid \tilde{a}) - q}{1-q}\, w'(x)\,\mathrm{d}x - \psi(\tilde{a}) \right]. \tag{IC}$$

I first consider a relaxed problem where (IC) is replaced by its first-order condition:

$$\int_{\underline{x}}^{x(a)} \frac{-F_a(x \mid a)}{1-q}\, w'(x)\,\mathrm{d}x = \psi'(a). \tag{FOA}$$

Since w is nondecreasing, the limited liability constraint is reduced to

$$w(\underline{x}) \geq 0. \tag{LL}$$

These constraints imply that $w(x)$ for $x \in (x(a), \overline{x}]$ are irrelevant for the agent's incentive. Given Assumption 1, it is optimal for the principal to set $w(x) = w(x(a))$ for all $x \in (x(a), \overline{x}]$. Thus,

[3] Since $\int_{\underline{x}}^{x(a)} \frac{f(x|a)}{1-q}\,\mathrm{d}x = 1$ and $F(x(a) \mid a) = 1 - q$, the integral in the second line can be interpreted as a corrected mean wage or the conditional expectation $\mathbb{E}[w(x) \mid x \leq x(a)]$.

the principal's cost minimization problem can be formulated as follows:

$$\begin{aligned}
&\min && \left[\int_{\underline{x}}^{x(a)} w(x) f(x \mid a)\,\mathrm{d}x + q w(x(a))\right] \\
&\text{subject to} && \int_{\underline{x}}^{x(a)} \frac{f(x \mid a)}{1-q}\, w(x)\,\mathrm{d}x - \psi(a) \geq U_0 && \text{(PC)} \\
& && \int_{\underline{x}}^{x(a)} \frac{-F_a(x \mid a)}{1-q}\, w'(x)\,\mathrm{d}x = \psi'(a) && \text{(FOA)} \\
& && w(\underline{x}) \geq 0 && \text{(LL)} \\
& && 0 \leq w'(x) \leq 1 \qquad \text{for a.e. } x && \text{(Lipschitz)}
\end{aligned}$$

Given (FOA) and (Lipschitz), the principal can induce only those actions a satisfying

$$\int_{\underline{x}}^{x(a)} \frac{-F_a(x \mid a)}{1-q}\,\mathrm{d}x \geq \psi'(a). \tag{3.1}$$

For action a such that (3.1) holds with equality, the solution is to make the agent residual claimant:

$$w^*(x) = \underline{w}^* + x - \underline{x},$$

where

$$\underline{w}^* = \max\left\{U_0 + \psi(a) - \int_{\underline{x}}^{x(a)} \frac{-F_a(x \mid a)}{1-q}\,\mathrm{d}x, 0\right\}.$$

In the remainder of the paper, I restrict attention to those actions satisfying (3.1) with a strict inequality:

$$A^* = \left\{a \in (0, \overline{a}) : \int_{\underline{x}}^{x(a)} \frac{-F_a(x \mid a)}{1-q}\,\mathrm{d}x > \psi'(a)\right\}$$

Define new state variables α and β by

$$\begin{aligned}
\alpha(x) &= \int_{\underline{x}}^{x} \frac{f(\tilde{x} \mid a)}{1-q}\, w(\tilde{x})\,\mathrm{d}\tilde{x}, & \alpha'(x) &= \frac{f(x \mid a)}{1-q}\, w(x), \\
\beta(x) &= \int_{\underline{x}}^{x} \frac{-F_a(\tilde{x} \mid a)}{1-q}\, w'(\tilde{x})\,\mathrm{d}\tilde{x}, & \beta'(x) &= \frac{-F_a(x \mid a)}{1-q}\, w'(x).
\end{aligned}$$

Then, the problem can be expressed as an optimal control problem with states w, α, and β and control w':

$$
\begin{aligned}
\min \quad & \left[\int_{\underline{x}}^{x(a)} w(x) f(x \mid a)\,\mathrm{d}x + q w(x(a))\right] \\
\text{subject to} \quad & w'(x) = u(x) \quad \text{for a.e. } x \\
& \alpha'(x) = \frac{f(x \mid a)}{1-q} w(x), \qquad \beta'(x) = \frac{-F_a(x \mid a)}{1-q} u(x) \\
& w(\underline{x}) \geq 0, \quad w(x(a)) \text{ free}, \\
& \alpha(\underline{x}) = 0, \quad \alpha(x(a)) - \psi(a) \geq U_0, \qquad \beta(\underline{x}) = 0, \quad \beta(x(a)) = \psi'(a) \\
& 0 \leq u(x) \leq 1 \quad \text{for all } x \in [\underline{x}, x(a)].
\end{aligned}
\tag{P}
$$

I define a class of simple contracts:

Definition 1. Wage schedule w is *(debt and) capped bonus* contract if there exist $y, z \in [\underline{x}, \overline{x}]$ with $y < z$ such that $w'(x) = 0$ if $x \in [\underline{x}, y) \cup (z, \overline{x}]$ and $w'(x) = 1$ if $x \in (y, z)$.

I show that, under the strict version of the monotone likelihood ratio property, the solution to (P) for each $a \in A^*$ is a debt and capped bonus contract.

Assumption 3 (Strict MLRP). $\frac{\partial}{\partial x}\left(\frac{f_a(x|a)}{f(x|a)}\right) > 0$ for all x.

Proposition 1. *Assume strict MLRP. For each $a \in A^*$, the optimal wage schedule w^* associated with the solution u^* to the relaxed problem* (P) *is a debt and capped bonus contract.*

Proof. See Appendix A. □

By the proof of Proposition 1, the optimal wage schedule for inducing $a \in A^*$ is either of the following three contracts:

- Capped bonus contract:

$$
w^*_{\mathrm{B}}(x) = \begin{cases} \underline{w^*_{\mathrm{B}}} + x - \underline{x} & \text{if } x \in [\underline{x}, z] \\ \underline{w^*_{\mathrm{B}}} + z - \underline{x} & \text{if } x \in (z, \overline{x}], \end{cases}
$$

where $\underline{w}^*_{\mathrm{B}}$ and z satisfies

$$\underline{w}^*_{\mathrm{B}} = U_0 + \psi(a) + \int_{\underline{x}}^{z} \frac{F(x \mid a)}{1-q}\,\mathrm{d}x - \left(z - \underline{x}\right) \geq 0$$
$$\int_{\underline{x}}^{z} \frac{-F_a(x \mid a)}{1-q}\,\mathrm{d}x = \psi'(a).$$

- Debt and capped bonus contract:

$$w^*_{\mathrm{DB1}}(x) = \begin{cases} 0 & \text{if } x \in \left[\underline{x}, y\right] \\ x - y & \text{if } x \in (y, x(a)] \\ x(a) - y & \text{if } x \in (x(a), \overline{x}], \end{cases}$$

where y satisfies

$$x(a) - y - \int_{y}^{x(a)} \frac{F(x \mid a)}{1-q}\,\mathrm{d}x - \psi(a) \geq U_0$$
$$\int_{y}^{x(a)} \frac{-F_a(x \mid a)}{1-q}\,\mathrm{d}x = \psi'(a).$$

- Another debt and capped bonus contract:

$$w^*_{\mathrm{DB2}}(x) = \begin{cases} 0 & \text{if } x \in \left[\underline{x}, y'\right] \\ x - y' & \text{if } x \in (y', z'] \\ z' - y' & \text{if } x \in (z', \overline{x}], \end{cases}$$

where y' and z' satisfies

$$z' - y' - \int_{y'}^{z'} \frac{F(x \mid a)}{1-q}\,\mathrm{d}x - \psi(a) = U_0$$
$$\int_{y'}^{z'} \frac{-F_a(x \mid a)}{1-q}\,\mathrm{d}x = \psi'(a).$$

Note that each contract has a constant wage region at the top. This comes from the dual risk-aversion; the agent does not consider the possibility of outcome levels higher than $x(a)$.

Note also that when (LL) binds, so for w^*_{DB1} and w^*_{DB2}, the wage contracts also have the form of *debt*. This is consistent with Innes (1990), as the agent is almost risk-neutral for outcome levels smaller than $x(a)$.[4]

I now turn to the validity of the first-order approach. Using the technique in Chade and Swinkels (2020), I derive a sufficient condition under which, given any of the above three contracts, the agent's net utility is strictly quasiconcave in action; thus, the (FOA) characterizes the optimal choice of action.

Proposition 2. *Under strict MLRP, the first-order approach is valid if*

$$\frac{F_{aa}(x \mid a)}{F_a(x \mid a)} < \frac{\psi''(a)}{\psi'(a)} \qquad \text{for all } x \in \big(\underline{x}, x(a)\big). \tag{3.2}$$

Proof. The proof is almost the same as Chade and Swinkels (2020, Proposition 5). Recall that the agent's net utility from wage schedule w and action $\tilde{a}$ is given by

$$U(w, \tilde{a}) = w\big(\underline{x}\big) + \int_{\underline{x}}^{x(\tilde{a})} \frac{1 - F(x \mid \tilde{a}) - q}{1 - q}\, w'(x)\, \mathrm{d}x - \psi(\tilde{a}).$$

Its first- and second-order derivatives with respect to action are

$$\begin{aligned} U_a(w, \tilde{a}) &= \int_{\underline{x}}^{x(\tilde{a})} \frac{-F_a(x \mid \tilde{a})}{1 - q}\, w'(x)\, \mathrm{d}x - \psi'(\tilde{a}) \\ U_{aa}(w, \tilde{a}) &= x'(\tilde{a})\, \frac{-F_a(x(\tilde{a}) \mid \tilde{a})}{1 - q}\, w'(x(\tilde{a})) - \int_{\underline{x}}^{x(\tilde{a})} \frac{F_{aa}(x \mid \tilde{a})}{1 - q}\, w'(x)\, \mathrm{d}x - \psi''(\tilde{a}), \end{aligned}$$

If $U_a(w, a) = 0$,

$$\psi''(a) = \frac{\psi''(a)}{\psi'(a)}\, \psi'(a) = \int_{\underline{x}}^{x(a)} \frac{\psi''(a)}{\psi'(a)} \left[\frac{-F_a(x \mid a)}{1 - q}\right] w'(x)\, \mathrm{d}x.$$

Using this equation, $U_{aa}(w, a)$ can be written as

$$U_{aa}(w, a) = x'(a)\, \frac{-F_a(x(a) \mid a)}{1 - q}\, w'(x(a)) - \int_{\underline{x}}^{x(a)} \frac{w'(x)}{1 - q} \left[F_{aa}(x \mid a) - \frac{\psi''(a)}{\psi'(a)} F_a(x \mid a)\right] \mathrm{d}x. \tag{3.3}$$

[4]The agent does not exhibit risk aversion in the sense that $\phi(1 - F(x))$ is linear for $x \le x(a)$.

Note that $U_{aa}(w,a) < 0$ suffices for $U(w,\cdot)$ to be strictly quasiconcave. For w^*_{B}, this condition is reduced to

$$-\int_{\underline{x}}^{z} \frac{1}{1-q}\left[F_{aa}(x \mid a) - \frac{\psi''(a)}{\psi'(a)}\, F_a(x \mid a)\right] \mathrm{d}x < 0.$$

This holds if

$$\frac{F_{aa}(x \mid a)}{F_a(x \mid a)} < \frac{\psi''(a)}{\psi'(a)} \qquad \text{for all } x \in \big(\underline{x}, z\big).$$

Similarly, for w^*_{DB2}, a sufficient condition for quasiconcavity is

$$-\int_{y'}^{z'} \frac{1}{1-q}\left[F_{aa}(x \mid a) - \frac{\psi''(a)}{\psi'(a)}\, F_a(x \mid a)\right] \mathrm{d}x < 0.$$

This holds if

$$\frac{F_{aa}(x \mid a)}{F_a(x \mid a)} < \frac{\psi''(a)}{\psi'(a)} \qquad \text{for all } x \in (y', z').$$

As w^*_{DB1} is not differentiable at $x(a)$, U_{aa} given by (3.3) is not well defined. However, I can evaluate the right derivative of $U_a\big(w^*_{\mathrm{DB1}}, \cdot\big)$ at $\tilde{a} = a$. It is given by

$$U^+_{aa}\big(w^*_{\mathrm{DB1}}, a\big) = -\int_{y}^{x(a)} \frac{1}{1-q}\left[F_{aa}(x \mid a) - \frac{\psi''(a)}{\psi'(a)} F_a(x \mid a)\right] \mathrm{d}x.$$

Hence, a sufficient condition for strict quasiconcavity is

$$\frac{F_{aa}(x \mid a)}{F_a(x \mid a)} < \frac{\psi''(a)}{\psi'(a)} \qquad \text{for all } x \in (y, x(a)).$$

Combining all the conditions, (3.2) is sufficient to validate the FOA. □

The condition (3.2) requires that for all $x \in \big(\underline{x}, x(a)\big)$, the outcome distribution F is (in proportionate terms) less convex in action than the disutility ψ. In particular, (3.2) is satisfied if both F and ψ are convex in action, as in Rogerson (1985).

4 Conclusion

This note has explored the implications of dual risk aversion in a standard moral hazard setting. Under this preference, the cost-minimizing contract for the principal takes a particularly simple form: a debt and capped bonus contract. This structural simplicity may facilitate further analysis of moral hazard problems in more complex or applied environments. Applying the present framework to a broader range of economic settings is a promising direction for future research.

A Proof of Proposition 1

Define the Hamiltonian for (P) as follows:

$$H(w, u, p_1, p_2, p_3, x) = -p_0 w f + p_1 u + p_2\left(\frac{f}{1-q}\, w\right) + p_3\left(\frac{-F_a}{1-q}\, u\right).$$

Let $u^*(x)$ be a piecewise continuous control defined on $\left[\underline{x}, x(a)\right]$ that solves problem (P) and let $w^*(x)$, $\alpha^*(x)$ and $\beta^*(x)$ be the associated optimal paths.[5] By Theorem 5 of Seierstad and Sydsaeter (1986, chap. 3), there exist constants p_0, p_2^*, and p_3^* and a continuous and piecewise continuously differentiable adjoint function $p_1(x)$ such that, for all $x \in \left[\underline{x}, x(a)\right]$,

$$\left(p_0, p_1(x), p_2^*, p_3^*\right) \neq (0,0,0,0) \tag{A.1}$$

$$p_0 = 0 \text{ or } p_0 = 1 \tag{A.2}$$

$$u^*(x) \text{ maximizes } H\left(w^*(x), u, p_1(x), p_2^*, p_3^*, x\right) \text{ for all } u \text{ in } [0,1] \tag{A.3}$$

$$\text{Except at the points of discontinuities of } u^*(x),\ p_1'(x) = p_0 f(x \mid a) - p_2^*\left[\frac{f(x \mid a)}{1-q}\right] \tag{A.4}$$

$$p_1(x(a)) = -p_0 q,\ \ p_2^* \geq 0 \text{ and } p_2^*[\alpha^*(x(a)) - \psi(a) - U_0] = 0 \tag{A.5}$$

$$p_1\left(\underline{x}\right) \leq 0 \text{ and } p_1\left(\underline{x}\right) w^*\left(\underline{x}\right) = 0 \tag{A.6}$$

Since the Hamiltonian is linear and hence concave in (w, u) for all $x \in \left[\underline{x}, x(a)\right]$, the maximized Hamiltonian $\widehat{H}(w, p_1, p_2, p_3, x) \equiv \max_{u\in[0,1]} H(w, u, p_1, p_2, p_3, x)$ is also concave in w for all $x \in \left[\underline{x}, x(a)\right]$. Thus, by Theorem 6 of Seierstad and Sydsaeter (1986, chap. 3), the necessary conditions (A.1) to (A.6) are also sufficient for optimality if they are satisfied with $p_0 = 1$.

[5] I choose the value of $u(x)$ at a point of discontinuity x' as the left-hand limit of $u(x)$ at x'. I also assume that $u(x)$ is continuous at $\underline{x}$ and $x(a)$.

By (A.4) and (A.5),

$$p_1(x) = \frac{p_2^*}{1-q}\,[1-q-F(x\mid a)] - p_0[1-F(x\mid a)] \Longrightarrow\; p_1(\underline{x}) = p_2^* - p_0.$$

Then,

$$\begin{aligned} & H\big(w,u,p_1(x),p_2^*,p_3^*,x\big) \\ =\; & -p_0 w f + \left[\frac{p_2^*}{1-q}\,(1-q-F) - p_0(1-F) + p_3^*\left(\frac{-F_a}{1-q}\right)\right]u + p_2^*\left(\frac{f}{1-q}\,w\right) \end{aligned}$$

and hence

$$\begin{aligned} H_u^*(x) \equiv\; & \frac{\partial H}{\partial u}\big(w^*(x),u,p_1(x),p_2^*,p_3^*,x\big) \\ =\; & \frac{p_2^*}{1-q}\,[1-q-F(x\mid a)] - p_0[1-F(x\mid a)] + p_3^*\left[\frac{-F_a(x\mid a)}{1-q}\right]. \end{aligned}$$

A necessary condition for (A.3) is

$$u^*(x) = \begin{cases} 1 & \text{if } H_u^*(x) > 0 \\ 0 & \text{if } H_u^*(x) < 0. \end{cases} \tag{A.7}$$

Suppose $p_0 = 0$. Then, $p_1(\underline{x}) = p_2^* \le 0$, so $p_2^* = 0$, which implies $p_1(x) = 0$ for all $x \in \big[\underline{x}, x(a)\big]$. By (A.1), $p_3^* \neq 0$. Since $H_u^*(x) = p_3^*\Big[\frac{-F_a(x|a)}{1-q}\Big]$, (A.7) implies either $u^*(x) = 1$ or $u^*(x) = 0$ for all $x \in \big[\underline{x}, x(a)\big]$. However, both of them violate $\beta^*(x(a)) = \psi'(a)$. Thus, $p_0 = 1$. Note $p_1(\underline{x}) = p_2^* - 1 \le 0 \Leftrightarrow p_2^* \le 1$.

Case 1: Suppose $p_2^* = p_3^* = 0$. Then, $H_u^*(x) = F(x \mid a) - 1 < 0$ for all $x \in \big[\underline{x}, x(a)\big]$. Hence, (A.7) implies $u^*(x) = 0$ for all $x \in \big[\underline{x}, x(a)\big]$, which violates $\beta^*(x(a)) = \psi'(a)$.

Case 2: Suppose $p_2^* > 0$ and $p_3^* = 0$. Then, for any $p_2^* \in (0,1]$, $H_u^*(x) = \frac{p_2^*}{1-q}[1-q-F(x\mid a)] - [1-F(x\mid a)] < 0$ for all $x \in \big(\underline{x}, x(a)\big]$. Hence, (A.7) implies $u^*(x) = 0$ for all $x \in \big[\underline{x}, x(a)\big]$, which violates $\beta^*(x(a)) = \psi'(a)$.

Case 3: Suppose $p_2^* = 0$ and $p_3^* \neq 0$. Then, $p_1(\underline{x}) = -1$, so $w^*(\underline{x}) = 0$. Since $H_u^*(x) =$

$-[1 - F(x \mid a)] + p_3^*\left[\frac{-F_a(x|a)}{1-q}\right]$, $H_u^*(\underline{x}) < 0$, and for $x \in (\underline{x}, x(a)]$,

$$H_u^*(x) > 0 \iff p_3^* > \frac{(1-q)[1 - F(x \mid a)]}{-F_a(x \mid a)} \equiv s(x).$$

Since $\lim_{x \to \underline{x}} s(x) = +\infty$ and the strict MLRP implies $s'(x) < 0$ for all $x \in (\underline{x}, \overline{x})$, there exists $y \in (\underline{x}, x(a))$ such that $p_3^* > s(x) \Leftrightarrow x \in [y, \overline{x}]$.[6] Then, (A.7) implies $u^*(x) = 0$ if $x \in [\underline{x}, y]$ and $u^*(x) = 1$ if $x \in (y, x(a)]$. The associated optimal path has the form of debt and capped bonus:

$$w^*(x) = \begin{cases} 0 & \text{if } x \in [\underline{x}, y] \\ x - y & \text{if } x \in (y, x(a)] \\ x(a) - y & \text{if } x \in (x(a), \overline{x}], \end{cases} \tag{A.9}$$

where y satisfies

$$\alpha^*(x(a)) - \psi(a) \geq U_0 \iff x(a) - y - \int_y^{x(a)} \frac{F(x \mid a)}{1-q}\,\mathrm{d}x - \psi(a) \geq U_0 \tag{A.10}$$

$$\beta^*(x(a)) = \psi'(a) \iff \int_y^{x(a)} \frac{-F_a(x \mid a)}{1-q}\,\mathrm{d}x = \psi'(a).$$

Case 4: Suppose $p_2^* > 0$ and $p_3^* \neq 0$. Then, $H_u^*(\underline{x}) < 0$, and for $x \in (\underline{x}, x(a)]$,

$$H_u^*(x) > 0 \iff p_3^* > \frac{(1-q)[1 - F(x \mid a)] - p_2^*[1 - q - F(x \mid a)]}{-F_a(x \mid a)} \equiv t(x).$$

[6] By the strict MLRP, for any $x < x'$ and $a < a'$,

$$\frac{f(x' \mid a')}{f(x' \mid a)} > \frac{f(x \mid a')}{f(x \mid a)} \iff f(x' \mid a')f(x \mid a) > f(x \mid a')f(x' \mid a). \tag{A.8}$$

Integrating both sides with respect to x' from x to $\overline{x}$, for all $x \in [\underline{x}, \overline{x})$,

$$\frac{f(x \mid a)}{1 - F(x \mid a)} > \frac{f(x \mid a')}{1 - F(x \mid a')},$$

which is equivalent to $f_a(x \mid a)[1 - F(x \mid a)] + f(x \mid a)F_a(x \mid a) < 0$ for all $x \in [\underline{x}, \overline{x})$ and a. Hence,

$$s'(x) = \frac{(1-q)\{f(x \mid a)F_a(x \mid a) + f_a(x \mid a)[1 - F(x \mid a)]\}}{[-F_a(x \mid a)]^2} < 0.$$

Note that $t(x) > 0$ for all $x \in \left(\underline{x}, x(a)\right]$ for any $p_2^* \leq 1$ and that

$$t'(x) = \frac{(1-q)[fF_a + (1-F)f_a] - p_2^*[fF_a + (1-q-F)f_a]}{(-F_a)^2}.$$

- If $p_2^* = 1$, then the strict MLRP implies $t'(x) > 0$ for all $x \in \left(\underline{x}, x(a)\right]$.[7] Hence, there exists $z \in \left(\underline{x}, x(a)\right)$ such that $p_3^* > t(x) \Leftrightarrow x \in \left[\underline{x}, z\right)$. Then, (A.7) implies $u^*(x) = 1$ if $x \in \left[\underline{x}, z\right]$ and $u^*(x) = 0$ if $x \in (z, x(a)]$. The associated optimal path has the form of capped bonus:

$$w^*(x) = \begin{cases} \underline{w}^* + x - \underline{x} & \text{if } x \in \left[\underline{x}, z\right] \\ \underline{w}^* + z - \underline{x} & \text{if } x \in (z, \overline{x}], \end{cases} \tag{A.11}$$

where $\underline{w}^*$ and z satisfies

$$\alpha^*(x(a)) - \psi(a) = U_0 \iff \underline{w}^* = U_0 + \psi(a) + \int_{\underline{x}}^{z} \frac{F(x \mid a)}{1-q}\,\mathrm{d}x - \left(z - \underline{x}\right) \geq 0$$
$$\beta^*(x(a)) = \psi'(a) \iff \int_{\underline{x}}^{z} \frac{-F_a(x \mid a)}{1-q}\,\mathrm{d}x = \psi'(a).$$

- If $p_2^* < 1$, then $p_1\left(\underline{x}\right) < 0$ and hence $w^*\left(\underline{x}\right) = 0$. Since the strict MLRP implies that $t(x)$ is a parabola opening upwards, there exist $y', z' \in \left(\underline{x}, x(a)\right]$ with $y' < z'$ such that $p_3^* > t(x) \Leftrightarrow x \in (y', z')$. Then, (A.7) implies $u^*(x) = 1$ if $x \in (y', z']$ and $u^*(x) = 0$ if $x \in \left[\underline{x}, y'\right] \cup (z', x(a)]$. The associated optimal path has the form of debt and capped bonus:

$$w^*(x) = \begin{cases} 0 & \text{if } x \in \left[\underline{x}, y'\right] \\ x - y' & \text{if } x \in (y', z'] \\ z' - y' & \text{if } x \in (z', \overline{x}], \end{cases} \tag{A.12}$$

[7]Note that
$$t'(x) = \frac{-q\{f(x \mid a)F_a(x \mid a) - F(x \mid a)f_a(x \mid a)\}}{[-F_a(x \mid a)]^2}.$$
Integrating (A.8) with respect to x from $\underline{x}$ to x', for all $x' \in \left(\underline{x}, \overline{x}\right]$,
$$\frac{f(x' \mid a')}{F(x' \mid a')} > \frac{f(x' \mid a)}{F(x' \mid a)},$$
which is equivalent to $f_a(x \mid a)F(x \mid a) - f(x \mid a)F_a(x \mid a) > 0$ for all $x \in \left(\underline{x}, \overline{x}\right]$ and a.

where y' and z' satisfies

$$\alpha^*(x(a)) - \psi(a) = U_0 \iff z' - y' - \int_{y'}^{z'} \frac{F(x \mid a)}{1-q} \, \mathrm{d}x - \psi(a) = U_0$$

$$\beta^*(x(a)) = \psi'(a) \iff \int_{y'}^{z'} \frac{-F_a(x \mid a)}{1-q} \, \mathrm{d}x = \psi'(a).$$

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